Subword Tokenizer & Prototype Training

Production-validated 35.2M-param TinyLM (0.0107 eval loss)

July 8, 2026

9. Project Roadmap: Physics Research Assistant SLM

9.1 Final Goal

Build a Physics Research Assistant using a 3B-7B parameter SLM, fine-tuned with LoRA on physics papers, combined with RAG and tool-using agents.

9.2 Current Status

Completed:
- Subword tokenizer (32K vocab, BPE)
- CLI for tokenization/detokenization
- Tokenizer validation (32K vs 50K trade-off analysis)
- Prototype training pipeline on Mac MPS
- Checkpointing and evaluation automation

Not Started:
- SLM model training pipeline (3B-7B parameters)
- LoRA fine-tuning setup
- RAG (Retrieval-Augmented Generation) integration
- Tool-using agent framework
- Physics paper dataset preparation at scale

9.3 Phase 1: Environment & Data Pipeline (Week 1-2)

Framework: Python 3.10+, PyTorch (with MPS), Hugging Face Transformers, Datasets, Accelerate, Pandas, NumPy

Steps:
1. Activate venv: source /Users/jdsingh/slm_v0/venv/bin/activate
2. Install packages: pip install torch transformers datasets accelerate peft
3. Verify MPS support for Mac
4. Download 10K-50K physics papers from arXiv
5. Organize into: /Users/jdsingh/slm_v0/data/physics_papers/raw/
6. Extract abstracts, introductions, methodologies
7. Remove noise, duplicates, non-English text
8. Normalize whitespace, split into chunks (max 4K words)
9. Tokenize using Rust tokenizer (32K vocab, frozen)
10. Split into train/val/test (98%/1%/1%), pack into 512-token sequences

Expected Outcome: 300M-1B tokens ready for SLM training.

9.4 Phase 2: Base SLM Training (Week 3-5)

Framework: PyTorch, Hugging Face Transformers, Accelerate

| Parameter | 3B Model | 7B Model |

|-----------|----------|----------|
Architecture	GPT2 or LLaMA-style decoder-only	GPT2 or LLaMA-style decoder-only
Hidden size	2048	3072
Layers	24	32
Attention heads	16	24
Max sequence length	512	512
Vocab size	32,000	32,000
Batch size	32	32
Gradient accumulation	4	4
Learning rate	5e-4 with cosine schedule	5e-4 with cosine schedule
Total steps	100,000	100,000

Success Criteria:
- Training loss decreases smoothly
- Validation perplexity < 20
- Generated samples coherent
- Peak memory < 16GB (Mac)

Expected Outcome: Production-ready base SLM (3B-7B parameters).

9.5 Phase 3: LoRA Fine-tuning (Week 6-7)

Framework: PEFT (Parameter-Efficient Fine-Tuning)

Parameter	Value
Rank	32
Alpha	32
Dropout	0.05
Target modules	q_proj, v_proj
Trainable parameters	~1-2% of base model

Training:
- Batch size: 16
- Learning rate: 2e-4
- Epochs: 3-5
- Output: LoRA weights (~50-100MB)

Expected Outcome: Fine-tuned SLM ready for RAG integration.

9.6 Phase 4: RAG Integration (Week 8-9)

Framework: FAISS, Sentence-Transformers, LangChain

Steps:
1. Generate embeddings for all physics paper chunks
2. Build FAISS vector index (384 or 768 dimensions)
3. Create retrieval function: query -> top-k relevant chunks
4. Augment prompt with retrieved context
5. Generate answer using fine-tuned SLM

Expected Outcome: RAG system with physics-grounded responses.

9.7 Phase 5: Tool-Using Agents (Week 10-11)

Framework: LangChain, LanGraph (ReAct framework)

Tool	Purpose
search_arxiv	Query arXiv for physics papers
solve_equation	Solve math equations (SymPy)
execute_python	Safe code execution for calculations

Agent Loop:
1. Generate thought + action
2. Execute appropriate tool
3. Observe result
4. Iterate until final answer
5. Max steps: 10-15

Expected Outcome: Multi-step reasoning with tool integration.

9.8 Phase 6: Deployment (Week 12+)

Framework: FastAPI, Docker

Endpoint	Method	Description
/query	POST	Submit physics question
/status	GET	Model health check
/upload_papers	POST	Add new physics papers to RAG

UI (Optional): Streamlit web interface

Deployment: Docker container on cloud (AWS, GCP, Azure, Hugging Face Spaces)

Expected Outcome: Production-ready Physics Research Assistant.

9.9 Timeline

Week	Phase	Deliverable
1-2	Data Pipeline	300M-1B tokenized tokens
3-5	Base SLM	3B-7B model checkpoint
6-7	LoRA	50MB LoRA weights
8-9	RAG	Vector index + retrieval
10-11	Agents	Multi-step reasoning demo
12+	Deployment	API + UI live

10 Hardware Requirements

Level	Specs
Minimum (M1/M2 Mac)	16GB RAM, 100GB SSD
Recommended	GPU with 24GB+ VRAM, 32-64GB RAM, 500GB NVMe SSD

| Training (AWS p4d.24xlarge) | 8x A100, 320GB total VRAM, spot instances |

10.1 Success Criteria

- Generates coherent physics-grounded answers
- Uses tools appropriately
- Cites retrieved papers
- Handles multi-step reasoning
- API response time < 5s
- Deployment-ready

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- `subword_tokenizer/src/lib.rs`: tokenizer core and model operations
- `subword_tokenizer/src/main.rs`: CLI dispatch (`bpe-tokenizer`)
- `stream_train.py`: prototype streaming training loop
- `scripts/evaluate_checkpoints.py`: checkpoint ranking
- `scripts/run_prototype_3x_and_select_best.sh`: train+select workflow
- `scripts/run_prototype_long_4h.sh`: long prototype preset

Architecture Diagram

Figure 1: Live arXiv Stream → Rust Tokenizer → TinyLM Training → Checkpoint Evaluation

```
Live arXiv Stream → stream_train.py → Retry/Backoff → Rust Tokenizer CLI → Token IDs → TinyLM Training → Checkpoints → Evaluation → best_checkpoint_*.json
```

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Updated July 5, 2026 — Added references and architecture diagram for IIT Bombay server request for end-of-day status handoff.

References

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